

Due by 6:00 PM 06/11. Web uploading.

Total 15 points as bonus points

- Compensation for the total class score will be made maximally up to 15 points as bonus point (i.e. 100+15 with bonus point 15).
- Total class score will be made with the scaling of the score obtained as following: midterm 30 points total, HW: 30 points total, final exam 30 points total, and 10 points for the class attendance.

Lienard-Wiechert potential, fields (3 points)

Start from the scalar and vector potentials in the Lorentz gauge,

$$\varphi(\vec{x}, t) = \int dt' \frac{d^3\vec{x}'}{|\vec{x} - \vec{x}'|} \rho(\vec{x}', t') \delta\left(t' + \frac{|\vec{x} - \vec{x}'|}{c} - t\right),$$

$$\vec{A}(\vec{x}, t) = \frac{1}{c} \int dt' \frac{d^3\vec{x}'}{|\vec{x} - \vec{x}'|} \vec{J}(\vec{x}', t') \delta\left(t' + \frac{|\vec{x} - \vec{x}'|}{c} - t\right),$$

with $\rho(\vec{x}, t) = e\delta[\vec{x} - \vec{r}(t)]$, $\vec{J}(\vec{x}, t) = ec\vec{\beta}\delta[\vec{x} - \vec{r}(t)]$, and $\frac{\vec{r}(t')}{dt} = c\vec{\beta}$,

1. (0.5) Obtain the Lienard-Wiechert potential, as in (14.18).

$$\varphi(\vec{x}, t) = \frac{e}{(1 - \hat{n} \cdot \vec{\beta})|\vec{x} - \vec{r}(t')|} \Bigg|_{retarded}$$

$$\vec{A}(\vec{x}, t) = \frac{e\vec{\beta}}{(1 - \hat{n} \cdot \vec{\beta})|\vec{x} - \vec{r}(t')|} \Bigg|_{retarded}$$

2. (0.5) From the definition of the retarded time, $t' = t - \frac{|\vec{x} - \vec{x}'|}{c}$, show that

$$dt' = -\frac{1}{c} \frac{\hat{n} \cdot d\vec{x}}{(1 - \hat{n} \cdot \vec{\beta})}, \text{ or } \nabla t' = -\frac{1}{c} \frac{\hat{n}}{(1 - \hat{n} \cdot \vec{\beta})}, \text{ for constant time (t) but with}$$

variation in the position.

3. (2) By letting $D = (1 - \hat{n} \cdot \vec{\beta})|\vec{x} - \vec{r}(t')|$, find the relations for $\frac{\partial D}{\partial t}$ and ∇D . Now, with these relations, find the E and B.

Note that

$$\vec{E}(\vec{x}, t) = \frac{e}{D^2} \left(\nabla D + \frac{\vec{\beta}}{c} \frac{\partial D}{\partial t} \right) - \frac{ec\dot{\vec{\beta}}}{D} \frac{\partial t'}{\partial t}$$

, and

$$\vec{B}(\vec{x}, t) = \frac{e}{D^2} \vec{\beta} \times \nabla D + \frac{e}{D} \nabla t' \times \dot{\vec{\beta}}$$

J.D. Jackson, Chapter 12 & 14 (3rd edition)

4. (3) Lorentz invariant Lagrangian (Prob. 12.1)
5. (3) Proca (photon with mass) Lagrangian (Prob. 12.16)
6. (3) Lienard-Wiechert fields (Prob. 14.4)
7. (3) Lienard-Wiechert fields for a simple harmonic field (Prob. 14.12)